

Fazle Rabbi Zaki

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Education

University of Michigan

Ann Arbor, MI

B.S.E. in Aerospace Engineering

May 2026

- University Honors (103rd Annual Honors Convocation); Dean's Honor List: Fall 2025, Winter 2026
- Coursework: Attitude Determination & Control Systems (grad), Engineering for Space Environment (grad), Automatic Control, Aircraft Dynamics, Space System Design, Spacecraft Dynamics, Propulsion, Aerodynamics

Technical Skills

GNC/ADCS:	Rigid-body attitude dynamics & kinematics, EKF design & consistency analysis, multi-sensor fusion, PID & bang-bang control, finite-state autonomy & mode logic, gyro-rate-feedback detumble, Monte Carlo validation
Software:	Python (NumPy, SciPy, pandas), MATLAB/Simulink, Simulink auto-code generation, C++, Arduino IDE, Git, Linux
Test & Integration:	HIL testbed design, SIL simulation, functional acceptance testing, sensor checkout, DAQ, oscilloscope, serial protocols (UART/I2C, CRC framing)
Systems Engineering:	Requirements flowdown & traceability, trade studies, mass/power/cost budgeting, interface definition, NASA Phase B PDR
CAD/Analysis:	SolidWorks, CATIA, STAR-CCM+ (CFD)
Hardware:	Raspberry Pi, Arduino, IMU (TDK ICM-20948), magnetometers, reaction wheels, magnetorquers, thermocouples

Engineering Projects

CubeSat ADCS Imaging Testbed

Ann Arbor, MI

ADCS & Systems Lead | University of Michigan (Graduate-Level Course)

Aug 2025 – Dec 2025

- Implemented a **2-state Extended Kalman Filter** (single-axis attitude and rate) in MATLAB/Simulink, fusing dual gyroscopes, dual magnetometers, a star tracker, and a sun sensor through **six sequential, validity-gated measurement updates** with Joseph-form covariance and an attitude-linearized magnetometer Jacobian
- **Validated estimator convergence and consistency by Monte Carlo simulation:** across randomized noise realizations, estimates converged from a 100° initial attitude error and stayed within their predicted σ (covariance) bounds
- Designed a **finite-state attitude controller** in Simulink autonomously sequencing detumble, coarse-point, fine-point, and imaging modes with **bidirectional state reversion**; the full mode sequence completed in ~ 50 s in closed-loop simulation, recovering autonomously from excursions
- Built and characterized a custom **Helmholtz-coil HIL testbed** ($\approx 168 \mu\text{T}$, over $3\times$ Earth's field, reproducible within 10%) with a rotating spin platform; in a battery-powered test, commanded reaction-wheel actuation **halted and reversed a ~ 1 rev/s spacecraft spin**, with gyro rates cross-verified against an independent hall-effect sensor
- Deployed the flight stack on a **Raspberry Pi** (auto-generated from Simulink) with a dual-IMU Arduino/C++ node streaming **CRC-16/CCITT-framed** sensor packets over serial, and authored a **Python/UDP ground station** rendering estimator states, actuator commands, and per-sensor health at 10 Hz
- Directed **requirements flowdown and traceability** (L1 to L3) across an 18-member team, defined cross-subsystem interfaces, and managed mass/power/cost budgets, delivering the integrated 0.96 kg CubeSat at \$3,000 against a \$6,000 cap

Icarus Heliophysics Mission

Ann Arbor, MI

Systems Engineering & ADCS | Safety, Policy & Risk Lead

Jan 2026 – Apr 2026

- Contributed to a **GNC sensor trade study** evaluating sun sensors, magnetometers, star trackers, and IMUs against accuracy, mass, power, and cost metrics for a **zero-momentum, three-axis stabilized architecture**
- Supported **requirements flowdown** from mission-level objectives to subsystem specifications across 10 subsystems; contributed to mass/power budgeting and interface definition through a **NASA Phase B PDR**
- Performed **safety and risk analysis:** mapped 12 rights-holders and identified 3 mission-critical risks for a dual-spacecraft heliophysics mission
- Ensured **regulatory compliance** with NPR 8715.6 (debris mitigation), ITAR/DDTC (export control), and FCC/ITU (spectrum allocation)

MASA Limelight – Liquid Bipropellant Rocket Avionics

Ann Arbor, MI

Avionics Subteam | Michigan Aeronautical Science Association

Aug 2025 – May 2026

- Built **hardware-in-the-loop (HIL) test infrastructure** for sensor and pyro telemetry verification on a **liquid bipropellant rocket** (LOx/RP-1, Phoenix engine) targeting 30,000 ft apogee, isolating telemetry-chain faults before hot-fire testing
- Wrote **embedded C++ firmware** for thermocouple-based thermal monitoring of propulsion components and debugged recovery-valve actuation circuits using an oscilloscope and DAQ
- Executed **cold-flow test operations:** configured ground support equipment (GSE) and ran E-Box sensor checkout prior to live propellant testing

Horizontal-Axis Wind Turbine Generator

Ann Arbor, MI

Design Engineer | University of Michigan

Jan 2024 – Apr 2024

- Designed and built a self-yawing 1.54 m horizontal-axis wind turbine on a \$110 budget; verified structural integrity via load testing (300 N lateral) and instrumented performance (tip-speed ratio up to 7.2, 6.45 V peak) with DAQ and oscilloscope